

Adjusting Amount of AI Explanation for Visual Tasks

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ABSTRACT

Explainable AI (XAI) has been designed to render AI comprehensible to humans by providing explanations of its processes. However, an excessive amount of explanation can lead to cognitive overload in users and the development of inappropriate trust in AI. To explore the appropriate amount of explanation, this study investigated the effects of explanation type (AI attention heatmap, AI goal, and AI reliability) on trust in XAI for a traditional visual search task in Experiment 1, and the effects of presenting adjusted explanations on task performance for an applied task in Experiment 2. As a result, displaying AI results alone increased trust and task performance for a low-complexity task, and displaying AI results with AI attention heatmaps that have high interpretability increased trust and task performance for a high-complexity task. This study demonstrated the importance of adjusting the amount of AI explanation to develop trust appropriately and improve task performance.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Empirical studies in HCI**;

KEYWORDS

XAI, Explanation, Trust, AI attention heatmap, Interpretability, AI goal, AI reliability, Visual search, Visual identification

ACM Reference Format:

Akihiro Maehigashi, Yosuke Fukuchi, and Seiji Yamada. 2024. Adjusting Amount of AI Explanation for Visual Tasks. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3650802>

1 INTRODUCTION

Artificial Intelligence (AI) has been widely used for visual tasks, including building damage assessment [27], security screening [6], and medical diagnosis [3, 40]. In these contexts, a user is required to decide to accept or reject a result provided by AI more quickly and accurately. However, effective human-AI cooperation remains a challenge.

Effectively engaging with AI requires users to adjust their trust and reliance on AI appropriately. Trust is defined as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability" [21]. Adaptive reliance on AI occurs when trust is adjusted to its actual reliability. This process is called trust calibration [21]. Adaptive trust calibration leads to maximizing task performance. However, the miscalibration of trust occurs [30].

In this context, explainable AI (XAI) is anticipated to play a crucial role in enhancing adaptive trust calibration and fostering effective collaboration between humans and AI. XAI has been designed to render AI comprehensible to humans by providing explanations of its processes [10]. Nevertheless, users frequently have difficulties understanding these explanations [2]. Also, an excessive amount of explanation can result in cognitive overload, and this leads to inappropriate trust calibration and task performance degradation [29]. Therefore, adjusting the amount of explanation is important for adaptively calibrating trust and enhancing task performance.

In this study, we conducted two experiments to investigate XAI with minimal and effective explanations that help users adaptively calibrate trust and increase task performance for visual tasks. In Experiment 1, we investigated the influence of three explanation types, that is, indicating the location that an AI focuses on with a heatmap (AI attention heatmap), the goal of AI development (AI goal), and the probability that an AI answers correctly (AI reliability), on user trust in XAI and the acceptance of its results, using a traditional visual search task. As a result, displaying AI results alone increased trust and acceptance for a low-complexity task, and displaying AI results with AI attention heatmaps that have high interpretability increased trust and acceptance for a high-complexity task.

On the basis of the findings in Experiment 1, we chose effective explanations for increasing trust and investigated the influence on task performance, using a practical visual identification task, in Experiment 2. As a result, displaying AI results alone increased task performance more than displaying them with explanations for a low-complexity task, and displaying AI results with AI attention heatmaps that have high interpretability increased task performance more than displaying them with additional explanations for a high-complexity task.

As a novel contribution of this study, we demonstrated the importance of adjusting the amount of explanation to develop trust and enhance task performance appropriately for visual tasks.

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650802>

2 RELATED WORK

2.1 Visual task with XAI

In the context of building damage assessment, XAI has been developed for post-disaster evaluation, identifying structural damages [16, 25]. Morrison et al. [27] qualitatively classified human users' strategies for the assessment and developed XAI assistants that visually frame target buildings with literal explanations according to the six types of assessment strategies. They experimentally compared the XAI assistants with those strategies and clarified the XAI assistant that constructed a causal argument to explain building damage showed the highest performance accuracy.

Moreover, regarding security screening, visualization methods for security screening have been developed using XAI, surrounding or coloring threatening items, such as guns, knives, and explosive devices [1]. Chavaillaz et al. [6] experimentally compared the effect of using AI that showed a red frame surrounding a threatening item (direct indication) with it surrounding a whole bag (indirect indication) for expert and novice screeners. As a result, novices showed better performance with AI than without it, while experts did not show any difference in their performance with and without AI. However, their study did not analyze the difference between using AI with direct and indirect indications from the viewpoint of AI explanations.

In the medical field, XAI assistants have recently been developed for medical imaging, such as X-ray images [2]. Security screening has the characteristic of visual search, but medical diagnosis in particular involves visual decision-making since it requires determining medical conditions and treatment options from medical images [3, 40]. Xie et al. [40] showed that XAI that displays key features of lesions on an X-ray image (e.g., by displaying highlights on the image with literal explanations) enabled physicians and radiologists to effectively understand diagnostic results from the XAI. Also, an XAI with a heatmap showing the area, shape, and size of a lesion on an X-ray image was demonstrated to increase clinicians' acceptance of its outputs and diagnostic accuracy and improve diagnostic time performance more than no AI assistant [3].

Furthermore, in relation to a user's cognitive load, Vasconcelos et al. [36] explored how visual explanations generated by XAI impact trust and acceptance, considering task difficulty and users' cognitive load induced by AI explanations. They conducted a maze task using XAI, where participants were presented with either literal explanations of directions (high cognitive load condition) or high-lighted directions on a maze (low cognitive load condition). The results revealed that participants exhibited reduced over-reliance and higher task performance when using XAI in the low cognitive load condition compared to the high one, regardless of task difficulty. This study suggested that AI explanations inducing high cognitive load could lead to over-trust and misuse of XAI.

These previous studies above developed XAI interfaces for domain-specific situations, stating that trust in AI is crucial for task performance. However, there are not that many studies directly investigating the effect of explanations on trust in XAI and task performance. In contrast, in this study, we experimentally show the general effects of adjusting the amount of AI explanation on trust and task performance for visual tasks.

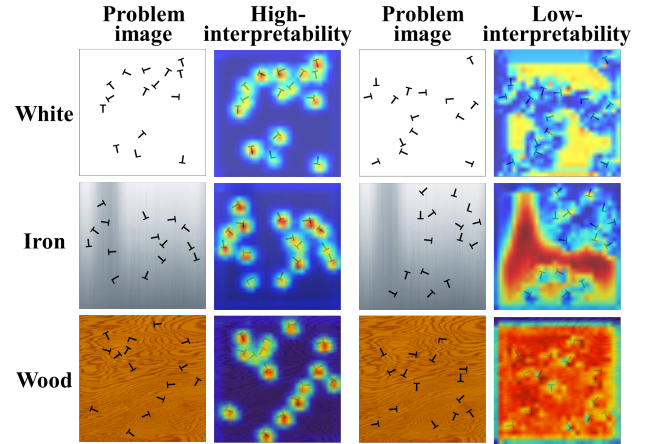


Figure 1: Problem images and high- and low-interpretability heatmaps for 15 stimuli with 3 types of background. All the images contain targets.

2.2 Type of explanation

There are multiple types of AI explanations. This study focused on three types, AI attention heatmap, AI goal, and AI reliability, since they are widely acknowledged and used explanations.

Regarding AI attention heatmaps, one widely adopted approach in XAI is Grad-CAM [5, 33]. Grad-CAM analyzes convolutional neural network (CNN) models to generate saliency maps (heatmaps) that highlight important features of visual images. However, it is difficult for humans to understand AI processes through its heatmaps. Therefore, recent studies have investigated how interpretable AI attention heatmaps could be displayed for humans to easily understand [15, 32].

Concerning AI goals, an AI indicating its goal increases the transparency and understandability of the AI and improves user trust [28]. The AI goal indicates the purpose and functionality for which a system was developed [21]. Aligning the goal of an AI with the user's goal is a theme that is addressed in HCI research [22, 26, 39]. These studies show that when AI goals match user objectives, it increases the sense of partnership with AI.

Regarding AI reliability, users generally tend to bolster their trust in AI when it demonstrates high levels of performance and reduce their trust when AI shows errors [20, 38]. Users are sensitive to AI errors, and even a minor error caused by AI leads to extreme under-trust [8] or algorithm aversion [7]. An effective strategy for avoiding such trust miscalibration involves displaying clear and comprehensible AI reliability to users [8].

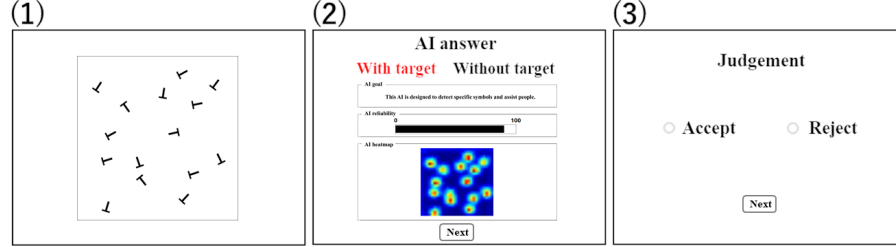
3 EXPERIMENT 1

3.1 Development of AI attention heatmaps

To develop heatmaps with high and low interpretability, we prepared a traditional visual search task where participants had to find the target "L" among distractors "T" [35]. We designed tasks involving 5 and 15 stimuli, each with or without the target. Also,

Table 1: Summarized results of pre-experiment 1. Since AI reliability was determined for each problem, there was no deviation among participants.

N of stimulus	Interpretability	AI reliability	Interpretability score	Human accuracy	Reaction time (ms)
5 stimuli	High	0.67	3.63 ($SD = 1.01$)	0.97 ($SD = 0.11$)	2721.14 ($SD = 910.01$)
	Low	0.60	2.26 ($SD = 0.79$)	0.96 ($SD = 0.12$)	2850.59 ($SD = 1133.90$)
15 stimuli	High	0.57	3.74 ($SD = 0.98$)	0.96 ($SD = 0.12$)	3917.72 ($SD = 2000.90$)
	Low	0.52	2.14 ($SD = 0.89$)	0.96 ($SD = 0.11$)	4189.93 ($SD = 3563.60$)

**Figure 2: Task procedure in Experiment 1. (1) Problem image was displayed at center of display for 2 seconds. (2) AI showed its answer in red with or without AI goal, AI attention heatmap, and AI reliability. (3) Participant decided to accept or reject answer by clicking. Participants repeated this procedure 30 times in this experiment.**

to develop a variety of types of heatmaps, different types of background, white, iron, and wood, were used.

Next, we trained a CNN-based classification model using the PyTorch library [31] and generated heatmaps. To implement the model, we used ResNet-18 [11] from the Torchvision package [23]. The accuracy was controlled to be in a range between 0.6 and 0.7. We compared heatmaps of the classification models generated from methods obtained from Grad-CAM, available in the pytorch-grad-cam package [9], and used Score-CAM for generating heatmaps [37]. To acquire low-interpretability heatmaps, we biased the training dataset and made an overfitted model. Figure 1 shows examples of problem images and high- and low-interpretability heatmaps.

Using the developed heatmaps, we conducted pre-experiment 1 with 200 participants to evaluate the interpretability of the heatmaps. To evaluate interpretability, an item related the explanation understandability from the explanation satisfaction scale [14] was used. On the basis of the evaluation results, the mean interpretability score of each heatmap was calculated. As a result, 30 high- and 30 low-scored heatmaps were selected for the both 5- and 15-stimulus tasks. In addition, we statistically confirmed that the interpretability score was higher for high-interpretability heatmaps than for low ones. Also, the reaction time was longer for the 15-stimulus task than the 5-stimulus one, meaning that the 15-stimulus task is more difficult than the 5-stimulus one. Table 1 shows the summarized results of pre-experiment 1.

3.2 Experimental task

This experiment was conducted with the problem images and heatmaps selected in pre-experiment 1 to investigate the influence of explanation type (AI attention heatmap, AI goal, and AI reliability) on trust in XAI and acceptance of its results. The experimental factors were the heatmap interpretability (high-interpretability

heatmap, low-interpretability heatmap, and no heatmap), AI goal (with and without goal indication), and AI reliability (with and without reliability indication) for the 5- and 15-stimulus tasks. Therefore, there was a total of 12 conditions for each type of task. The task procedure is shown in Figure 2. In addition, the length of time for displaying a problem image, 2 seconds, was decided on the basis of the results of pre-experiment 1 to highlight the task complexity between the 5- and 15-stimulus tasks.

3.3 Method

A total of 100 participants were recruited through a cloud-sourcing service provided by Yahoo! Japan. They were randomly assigned to one of the conditions and conducted a task. However, 13 participants were detected as inattentive by the Directed Questions Scale (DQS) [24] and were excluded from the analysis. As a result, data of 87 participants (64 male and 23 female from 26 to 65 y/o, $M = 48.70$, $SD = 9.54$) were used. Each participant answered regarding the original images 30 times with 5 or 15 stimuli. The order of the images was randomized. After finishing 30 problems, they answered on a trust scale for XAI [14] consisting of 8 items on a 5-point scale (1: I agree strongly - 5: I disagree strongly). After that, they started the task with 15 or 5 stimuli. The order of the number of stimuli was counterbalanced between participants.

3.4 Results and discussion

To investigate the effect of each explanation on trust and acceptance, we performed structural equation modeling (SEM) analyses. First, data sets were prepared based on the variables for each participant. For the heatmap interpretability, the interpretability score rated in pre-experiment 1 was used as a representative value of each heatmap. For the no-interpretability condition, "0" was used since there was no interpretability. Also, for the AI goal indication, the

Table 2: Results of SEM models for 5 and 15 stimuli. Table shows dependent and independent variables and estimated value (Estimate), standard error of estimated value (Std.Error), z-value, and p-value. * $p < 0.05$, ** $p < 0.01$, * $p < 0.001$**

N of stimulus	Dependant variable	Independent variable	Estimate	Std.Err	z-value	p-value
5 stimuli	Acceptance	Trust	0.07	0.02	4.435	*** < 0.001
	Trust	Heatmap interpretability	0.17	0.10	1.70	0.09
		AI goal	0.06	0.20	0.28	0.78
		AI reliability	0.07	0.19	0.34	0.74
		AI accuracy	3.08	1.14	2.69	** 0.01
15 stimuli	Acceptance	Trust	0.13	0.02	6.20	*** < 0.001
	Trust	Heatmap interpretability	0.21	0.09	2.50	* 0.01
		AI goal	-0.14	0.18	-0.77	0.44
		AI reliability	0.16	0.19	0.82	0.41
		AI accuracy	1.43	0.78	1.85	0.06

with- and without-goal indications were represented using “1” and “0” as dummy variables. Moreover, for AI reliability indication, the with- and without-reliability indications were also represented using “1” and “0.” Finally, the trust scores rated after 30 problems were used for the trust variable, and the acceptance rate (the rate at which the participants accepted AI results) was calculated.

Next, we developed SEM models for the 5- and 15-stimulus tasks. However, these models did not fit the data well. Therefore, we modified the models by adding AI accuracy, the mean accuracy rate of AI, as a variable to determine the path to trust. The modified models fit the data well, and the goodness-of-fit values settled within the permissible range [17]. Statistical power analyses with R software with alpha at .05 revealed that the SEM, performed with sample sizes ($N = 87$ for 5- and 15-stimulus tasks) for exact-fit tests, obtained powers of .96, and .97, showing satisfactory statistical powers.

As revealed by the SEM analyses (Table 2), while heatmap interpretability positively impacted trust in the 15-stimulus task, this effect was not found in the 5-stimulus task. Also, no influence of AI goal indication on trust was found. Moreover, there was no influence found from AI reliability indication on trust. Notably, AI accuracy exhibited a positive influence on trust in the 5-stimulus task, but this effect was not observed in the 15-stimulus task, meaning that participants acknowledged AI accuracy based on its presented results in the 5-stimulus task.

These results are consistent with the previous study showing that users’ perceived and actual reliance on AI and their eye fixation time on explanations of AI increased when the task was more difficult for users [4].

On the basis of the results of Experiment 1, we focused on the impact of explanations on user task performance for an applied visual identification task in Experiment 2.

4 EXPERIMENT 2

4.1 Development of AI attention heatmaps

In Experiment 2, we used only high-interpretability heatmaps, which increased trust and acceptance in Experiment 1, to investigate whether they enhance task performance in an applied task. First, we prepared an applied visual identification task: the strawberry task, in which the participants were required to identify a rotten strawberry as a target among fresh ones as the distractors.

The strawberry images were used from a dataset [34]. The task aligns with critical domains such as food safety. Next, we generated the heatmaps in the same manner as we did in the pre-experiment 1 (Figure 3). As a result, there was a total of 100 heatmaps prepared for interpretability rating.

Using the developed heatmaps, we conducted pre-experiment 2 with 200 participants to evaluate the interpretability of the heatmaps as in pre-experiment 1. As a result, 30 high-scored heatmaps were selected for the both 5- and 15-stimulus tasks. In addition, we statistically confirmed that the reaction time was longer for 15 stimuli than 5 stimuli, meaning that the 15-stimulus task is more difficult than the 5-stimulus one. Moreover, we confirmed that the interpretability score was higher than the middle rating score. Table 3 shows the summarized results of pre-experiment 2.

4.2 Method

In Experiment 1, we found positive influences of AI accuracy on trust for the 5-stimulus task and from the heatmap interpretability on trust for the 15-stimulus task. Therefore, in Experiment 2, we set up three experimental conditions: the no-explanation condition (showing only AI results), the AI-attention-heatmap condition (showing AI results with AI attention heatmaps that have high interpretability), and the all-explanation condition (showing AI results with AI attention heatmaps that have high interpretability, AI goal, and AI reliability) to compare the influences of AI explanations on task performance. We used the same task procedure as in Experiment 1, using the strawberry task, with 5 and 15 stimuli. Figure 4 shows the display of the AI result in each condition for the strawberry task.

A priori power analysis with G*Power indicated that 159 participants were needed for a medium effect size ($f = .25$) with the power at .80 and alpha at .05 for one-factor three conditions with a between-participant design. Therefore, a total of 200 participants were recruited through a cloud-sourcing service provided by Yahoo! Japan. They were randomly assigned to one of the conditions in one of the tasks and conducted a task. However, one participant was detected as inattentive by the attention check items of DQS and was excluded from the analysis. As a result, data of 199 participants (140 male and 59 female from 18 to 65 y/o, $M = 48.63$, $SD = 9.95$) were used. The procedure was basically identical to Experiment 1. However, in Experiment 2, each participant answered 30 problems

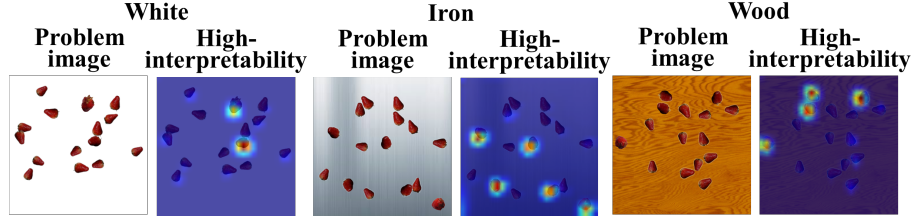


Figure 3: Problem images of 15 stimuli with three types of background and high-interpretability heatmaps. All the images contain targets.

Table 3: Summarized results of pre-experiment 2. Since AI reliability was determined for each problem, there was no deviation among participants.

N of stimulus	AI reliability	Interpretability score	Human accuracy	Reaction time (ms)
5 stimuli	0.70	3.75 ($SD = 0.61$)	0.84 ($SD = 0.14$)	4016.11 ($SD = 2048.30$)
15 stimuli	0.63	3.33 ($SD = 0.79$)	0.79 ($SD = 0.16$)	4447.78 ($SD = 2035.92$)

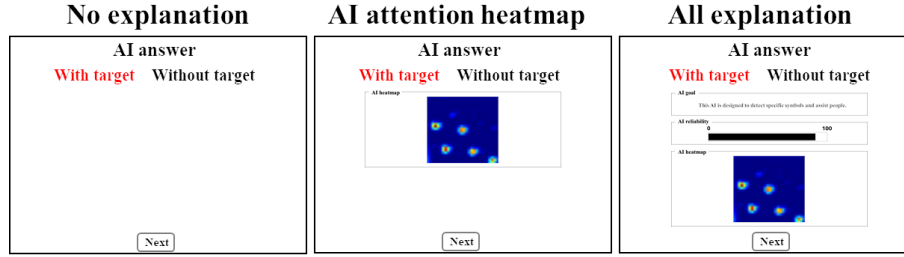


Figure 4: Displays for three experimental conditions for 15 stimuli in Experiment 2

for 5 and 15 stimuli in the strawberry task depending on the experimental condition. In addition, the participants were told that the target was a rotten strawberry, along with some images of rotten strawberries that were not used in the experimental task. Moreover, the length of displaying a problem image, 3 seconds, was decided based on the results of pre-experiment 2 to highlight the difference in task complexity between the 5- and 15-stimulus tasks.

4.3 Results and discussion

Figure 5 shows the mean task performance (the rate at which the participants accurately accepted or rejected AI results). As a result of an ANOVA on the task performance, regarding 5 stimuli, there was a significant main effect ($F(2, 196) = 4.55, p = .01, \eta_p^2 = .04$). The results of multiple comparisons with a Bonferroni test showed that the performance was higher in the no-explanation condition than the AI-attention-heatmap condition ($t(196) = 2.72, p = .01, r = .19$) and the all-explanation condition ($t(196) = 2.56, p = .02, r = .18$). There was no significant difference between the AI-attention-heatmap condition and the all-explanation condition ($t(196) = 0.14, p = .89, r = .01$). Regarding 15 stimuli, there was a significant main effect ($F(2, 196) = 3.69, p = .03, \eta_p^2 = .04$). The results of multiple comparisons with a Bonferroni test showed that the performance was higher in

the AI-attention-heatmap condition than the all-explanation condition ($t(196) = 2.55, p = .03, r = .18$). There was no significant difference between the no-explanation condition and the AI-attention-heatmap condition ($t(196) = 2.16, p = .06, r = .15$) and the no-explanation condition and the all-explanation condition ($t(196) = 0.43, p = .67, r = .05$).

For 5 stimuli, participants in the no-explanation condition exhibited a better task performance compared with those in the AI-attention-heatmap and all-explanation conditions. This result was confirmed because overloading users with AI explanations was considered to lead to cognitive overload and hinder performance as shown in a previous study [29]. Moreover, for 15 stimuli, the high-interpretability heatmaps in the AI-attention-heatmap condition significantly outperformed the all-explanation condition. While high-interpretability heatmaps have been shown to aid users in identifying crucial information in visual tasks [32], they particularly enhance task performance in highly complex tasks.

5 GENERAL DISCUSSION

For the low-complexity task, presenting AI results alone increased trust and acceptance, while the absence of explanations enhanced user task performance. In contrast, for the high-complexity task, presenting AI results along with the AI attention heatmap with high interpretability increased trust, acceptance, and notably improved

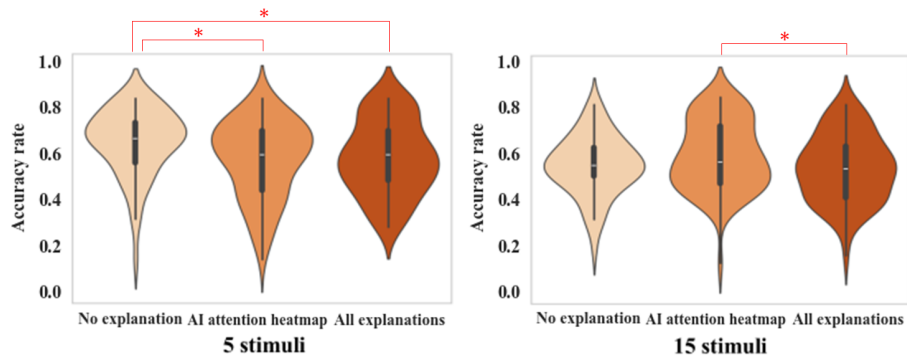


Figure 5: Task performance with 5 and 15 stimuli. * $p < 0.05$

user task performance compared with the presentation of all three explanations. This study highlights the importance of presenting minimal and optimal explanations for different complexity tasks. In other words, the negative impact of overloading users with explanations was observed as shown in the previous studies [15, 18, 29].

As a limitation of this study, the findings are primarily focused on visual search and identification tasks. Their direct applicability to other types of visual tasks, such as medical diagnosis where clinicians determine medical conditions and treatment options by interpreting medical images [3, 40]. Additionally, this study focused on a specific set of explanations, and there could be other effective explanations that were not explored. Furthermore, presenting representations with identical information in different formats would lead to variations in information processing costs [12, 13, 19]. Therefore, investigating the impact of displaying explanations in different forms becomes an important avenue for future research.

6 CONCLUSION

This study aimed to improve AI design for visual tasks and optimize the amount of explanation provided to users through two experiments. In the first experiment, presenting AI results for low-complexity task scenarios and including AI attention heatmaps with high interpretability for high-complexity task scenarios increased user trust and acceptance. The second experiment confirmed that presenting AI results alone improved user task performance in low-complexity task scenarios, while displaying AI attention heatmaps improved user task performance in high-complexity task scenarios. The results highlight the critical role of presenting optimal explanations in reducing cognitive load while improving user trust and task performance.

ACKNOWLEDGMENTS

This work was (partially) supported by JST, CREST (JPMJCR21D4), Japan.

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